On Detection of Diabetic Retinopathy via Multiple Instance Learning

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ABSTRACT

Diabetic Retinopathy (DR) is a complication of diabetes, caused by a damage to the blood vessels in the light-sensitive tissue of the retina. Since it affects the eyes, it can determine visual impairment or even blindness. Considering the number of diabetic patients worldwide, it is clear that effective screening of potential DR patients is of utmost importance. While direct and indirect ophthalmoscopy are the main methods for evaluating DR, artificial intelligence is on the rise in vision care. DR is detectable by analyzing data from patients' fundus photographs, and is therefore a disease that artificial intelligence tools can effectively support. In this paper, we present some preliminary numerical results obtained in discriminating between eye fundi of healthy individuals and of people with severe diabetic retinopathy, by using a Multiple Instance Learning approach.

KEYWORDS

Multiple Instance Learning, Diabetic Retinopathy Detection, Image Processing,

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1 INTRODUCTION

Artificial intelligence (AI) frameworks utilize advanced algorithms to face the screening of specific pathologies analyzing vast amounts of clinical data, with the ultimate goal of achieving tasks with minimal human involvement. AI is undoubtedly a major frontier in the healthcare domain, despite the complex and time-consuming task of extracting knowledge from high-dimensional datasets, including medical images. Solutions for data analysis and image analysis, based on Machine Learning (ML) and AI, have been provided in the literature and are extensively used as medical information tools for data prediction and diagnostic procedures [33], as well as in applications for specific lesion characterization and feature detection [28].

In particular, in this paper we focus on the detection of the Diabetic Retinopathy (DR) by image analysis of the eye fundus, using a Multiple Instance Learning (MIL) technique. Diabetes is a chronic disease directly related to the production of insulin, a hormone that regulates blood sugar levels. If the pancreas is unable to produce enough insulin, diabetes is referred to as *type 1*, while if the body is unable to effectively use the produced insulin, diabetes is referred to as *type 2*. Regular screening and early diagnosis are essential for all diabetes patients as they have the potential to develop Diabetic Retinopathy (DR). DR is a common, but serious complication of diabetes that affects the eyes, and it is the primary cause of new cases of blindness among people aged 20-74. It affects approximately 30% of diabetes patients, and is currently the leading cause of preventable blindness in the working-age population. DR occurs when the blood vessels in the retina, the light-sensitive tissue at the back of the eye, are damaged due to diabetes. Anyone with type 1 or type 2 diabetes can develop DR, and the longer a person has diabetes, the higher their risk of developing this eye complication. At the time of their first diabetes diagnosis, up to 21% of patients with type 2 diabetes also present retinopathy. In many cases, people with diabetic retinopathy may not notice symptoms until significant damage has already occurred. Therefore, it is straightforward that early diagnosis of DR is critical, as it leads to a better response to treatment and a more positive prognosis. Screening can be carried out through fundus examination or retinography, with the latter being more commonly recommended and forming the basis of current DR screening protocols. All diabetes and ophthalmology associations currently recommend annual retinography screening for all diabetes patients from the first diagnosis. Screening programs are usually carried out at ophthalmology units, where a patient's retinas can be evaluated by an ophthalmologist.

Telemedicine systems are available in developed countries, where patients can have their retinas photographed and examined by

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healthcare professionals. However, this is not considered cost-effective as it requires a large number of specialists to grade each patient's retinas manually. In addition, current screening services cannot keep up with the demand, with fewer than 50% of diabetes patients receiving regular examinations. To address this issue, several automatic reading systems are being developed that use artificial intelligence-based algorithms to detect DR. These algorithms have performed well in various DR classification tasks and are cost-effective compared to manual grading, which is the current gold standard. Most related works provide binary classification, distinguishing patients with more than mild DR from those without DR.

Unfortunately, diabetes is on the rise worldwide. The International Diabetes Federation (IDF) estimates that there are currently 463 million people worldwide suffering from diabetes mellitus (DM); moreover the IDF report [20] shows that the number of diabetes cases has significantly increased in recent decades, with projections indicating that there could be 600 million people with diabetes by 2035 and 700 million by 2045. Figure 1 shows estimated share of people 20-79 y/o with diabetes worldwide in 2019 and 2045.

Given the large number of diabetes patients worldwide, effectively screening potential DR cases requires a significant amount of resources. In the United States alone, universal diabetic retinopathy screening would require evaluating 32 million retinal images annually. However, early detection and intervention can prevent most vision loss caused by diabetic retinopathy. Therefore, there is a need to develop tools that allow for early and accurate diagnosis, which can assist specialists and enable the release of low-cost solutions for effective self-diagnosis. While direct and indirect ophthalmoscopy is the primary method for evaluating diabetic retinopathy, AI and big data have become increasingly popular in the eye care sector. These tools provide low-cost, effective, and potentially powerful solutions to support early and accurate diagnosis, enabling specialists to determine specific treatments.

The paper is organized as follows. In the next section, we highlight which are the visual features that are indicative of the presence of DR pathology. In Section 3 we describe the MIL paradigm and the related technique used for our preliminary experiments, presented in Section 4. Finally, in Section 5 some conclusions are drawn, accompanied by some considerations on possible future research directions.

2 DIABETIC RETINOPATHY DETECTION VIA ARTIFICIAL INTELLIGENCE

Like all other tissues in the human body, the retina is supplied with blood through the body's vasculature. Specifically, micro blood vessels continuously supply blood to the retina, which maintains the blood sugar level. However, high levels of sugar in the blood can cause the tiny blood vessels that nourish the retina to become blocked, cutting off its blood supply. In response, the eye attempts to grow additional blood vessels that do not develop properly and can easily leak.

The most common scale used by clinicians to grade DR is the International Clinical Diabetic Retinopathy (ICDR) scale, which has five grades: no DR, mild DR, moderate DR, severe DR, and proliferative DR. Each stage of the disease is associated with specific features. These features include microaneurysms, exudates, hemorrhages, abnormal blood vessels, and neovascularization. The presence or occurrence of one, two, or all of these characteristics in the retina determines the stage of DR [18].

In the early stages of diabetic retinopathy, patients may not experience any noticeable symptoms. In fact, the number of undiagnosed diabetic cases (Figure 2) is in general high. However, as the condition progresses, patients may begin to experience a range of symptoms, including distorted or blurred and fluctuating vision, dark or empty areas in their vision, spots or dark strings that float in their vision (known as floaters), and a progressive loss of visual acuity.

Figure 3 reports the same scene seen by normal eyes and eyes affected by DR, whose grades can be categorized into non-proliferative DR (NPDR) and proliferative DR (PDR).

Figure 4 depicts a fundus photograph of a person's left eye with mild (early) nonproliferative diabetic retinopathy and centre-involving diabetic macular edema (DME). The image highlights small red "dots" (microaneurysms) and "blots" (hemorrhages) with typical examples marked by white arrowheads. The figure shows retinal exudates, several of which are within 500 microns of the centre of the fovea. The yellow circle outlines an area of central thickening of the retina (edema).

Figure 5 is a diagrammatic representation of an advanced proliferative diabetic retinopathy with ischemic maculopathy. The image shows laser scars resulting from panretinal laser photocoagulation, including areas of focal grid laser treatment within the macula. The extensive capillary dropout seen in the image is indicative of end-stage diabetic retinopathy. While additional laser treatment may help to arrest the proliferative changes, very little vision remains due to the obliteration of the inner retinal vasculature.

The standard features together with the corresponding DR type are reported in Table 1.

Feature	DR Type
small microaneurysms	Mild NPDR
hemorrhages, exudates, microaneurysms	Moderate NPDR
retinal ischemias and abnormal features	Severe NPDR
abnormal blood vessels, all abnormal features	PDR

Table 1: Standard Features of DR

For individuals with uncontrolled diabetes, it is recommended to undergo annual fundus screening to facilitate early identification and treatment of diabetic retinopathy, which can help to decrease the disease burden in the community. However, only half of these patients receive regular screening due to various factors such as lack of access to an ophthalmologist in their area, logistical challenges, or the high cost of the visit. To overcome these challenges, there has been an increased interest in using completely automated AI-based grading systems for grading pictures. Such technology would enable real-time determination of whether a patient requires referral, and it may be much less expensive compared to having ophthalmologists conduct the screening.

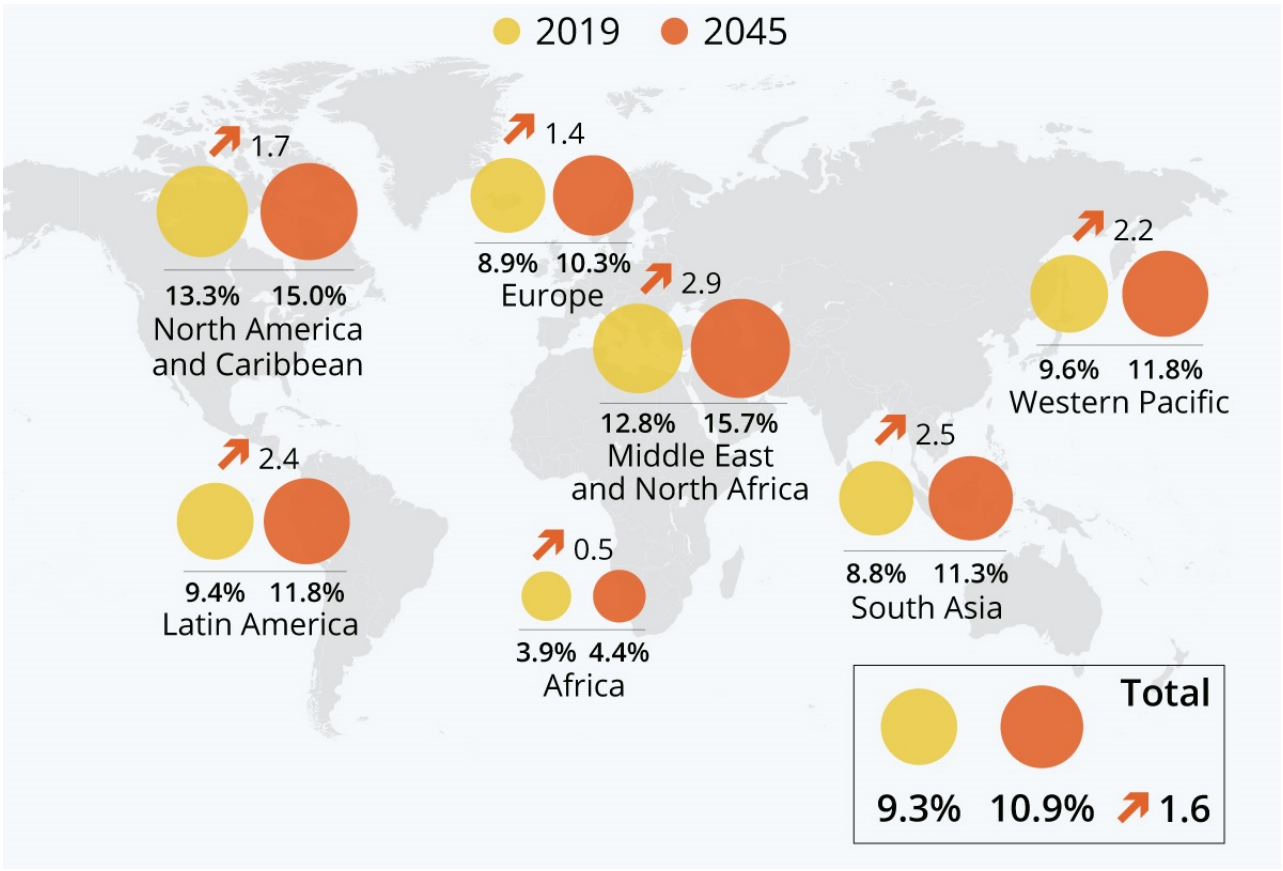


Figure 1: Estimated distribution of people, aged 20-79, with diabetes worldwide in 2019 and 2045 (in percent) from IDF [20]

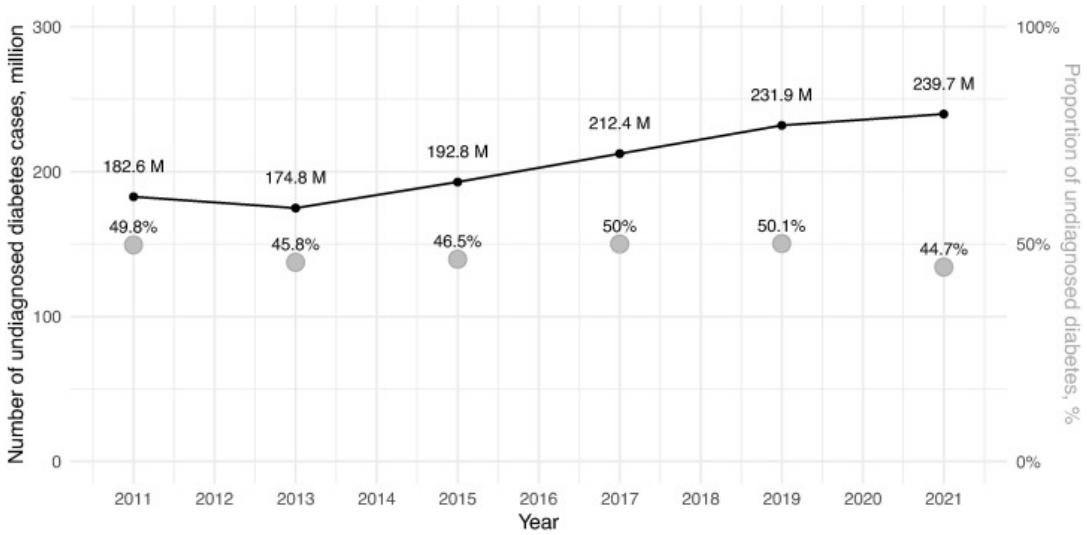


Figure 2: Number of undiagnosed diabetic cases. Source IDF 10th Ed. 2021

In April 2018, the US Food and Drug Administration (FDA) certified an AI algorithm created by IDX Systems Corporation (IDX) [1] for use with the Topcon Fundus camera to detect diabetic retinopathy.

[1]IDX was a healthcare software technology company founded in 1969 acquired by General Electric and incorporated into its GE Healthcare business unit in 2006.



Figure 3: The different vision of the same scene

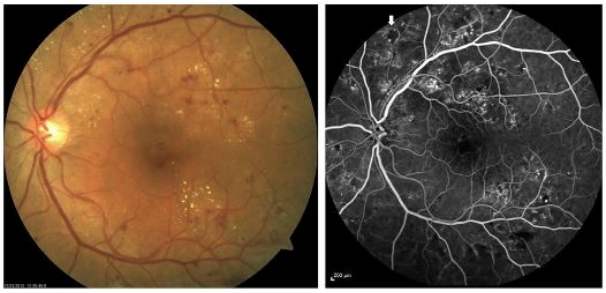


Figure 4: Non-proliferative DR (NPDR)

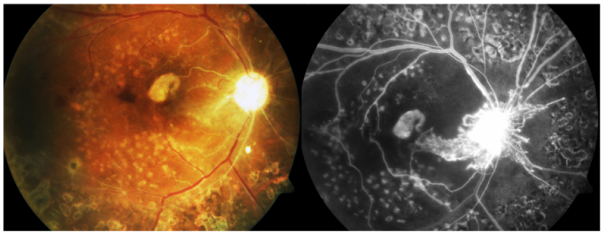


Figure 5: Proliferative DR (PDR)

Several studies have been conducted to develop AI-based algorithms for detecting DR, including models based on microaneurysms and hemorrhages identification by Wong et al. [30] and an approach for detecting exudation and blood vessels using morphological component analysis by Imani et al. [17]. Adaptive thresholding is used to create the vessel map. Using inverse surface thresholding, Yazid et al. [29] were able to identify hard exudation and the optic disc.

Ting et al. [25] conducted a significant research in Singapore using numerous retinal pictures to validate Deep Learning (DL) and found that it has a high sensitivity and specificity in detecting DR and other eye disorders. Using DL algorithms, recent studies have achieved remarkable results, thus demonstrating the effectiveness of these techniques in automatic diagnosis systems. However, the problem of the interpretability of the results by specialists arises. The phisicians express reservations about solutions that work like a "black box" and whose diagnoses are not based on the features arising from the medical protocols of reference, but end up to be seen as mathematical-computer alchemy not always well interpretable.

There has been a lot of research focused on developing Explainable AI in the medical field, with a particular focus on diagnosis, decision support, and interpretability. In this field, interpretability is not just a matter of intellectual curiosity, but is a critical factor due to the significant impact medical choices have on patients' lives, and the associated risks and responsibilities that clinicians bear [9].

The current solutions, based on AI and DL, leave open some important questions relating to better interpretability, accuracy and robustness even in the presence of non-annotated datasets or with erroneous/incomplete annotations. The way is paving for the adoption of hybrid frameworks, which use DL algorithms for the preprocessing [23] and segmentation steps of the medical images [22, 24] and then adopt advanced ML algorithms and heuristics [4, 13, 27] for the classification steps. The adoptable features can therefore also be defined by specialists and not just already extracted from a system that works like a "black box". To this end, the MIL paradigm makes it possible to detect local patterns of interest in images and videos, based exclusively on globally assigned class labels. The classification performances can also give important results using ensemble learning configurations [12, 31]. In this sense, what is presented in this work is meant to be functional for the creation of a hybrid solutions.

3 A MULTIPLE INSTANCE LEARNING APPROACH

Multiple Instance Learning (MIL) [16] is a classification technique whose purpose is to discriminate between sets of points. In the MIL terminology, the sets are called bags and the points inside them are called instances. The main characterization of a MIL approach with respect to a classical supervised classification technique resides in the fact that the learning phase is performed by exploiting only the class labels of the entire bags, since the class label of each single instance is unknown.

The MIL paradigm has been introduced for the first time in [11], where the objective was to classify drug molecules (each of them represented by a bag) as active or non-active, in function of the possible three-dimensional conformations (instances) they can assume.

MIL is applied in various contexts such as image classification, bankruptcy prediction, text classification, speaker identification and so on. More specifically, it plays a relevant role in medical image and video analysis (see [21]), and especially in diagnostics by images analysis, which is a fundamental field supporting physicians to have early diagnoses.

The most common MIL problem treated in the literature is the binary case (i.e. two classes of bags), where it is possible to have also more than two classes of instances. For such problems, there are mainly three types of approaches (see for example [1, 8]). The first one is the bag-level approach, where each bag is treated as a global entity, while the second one is the instance-level approach (see for example [5, 7, 15, 19]) where the classification is performed in the instance space, obtaining the bag label as aggregation of the labels of the corresponding instances. The third type of approach (see for example [2]) is a compromise between the two previous ones and consists in representing each or some bag by one of its instances, involved successively in the classification process.

The MIL technique that we adopt in this work for diabetic retinopathy (by means of image detection) is the mi-SPSVM (multiple instance - Semi-Proximal Support Vector Machine) algorithm [5], which is an instance-level approach solving binary MIL problems characterized by two classes of instances. In the literature, such kind of MIL problems are based on the so-called standard MIL assumption, stating that a bag is considered positive if it contains at least a positive instance and it is negative otherwise.

Algorithm mi-SPSVM has been introduced in [5] and it exploits the good properties exhibited for supervised classification by the SVM (Support Vector Machine) technique in terms of accuracy and by the PSVM (Proximal Support Vector Machine) approach [14] in terms of efficiency. It computes a separating hyperplane

$$H(w,b) \triangleq \{x \in \mathbb{R}^n \mid w^T x + b = 0\}, \tag{1}$$

solving, at each iteration, the following quadratic optimization problem:

$$\begin{cases} \min\limits_{w,b,\xi} & \frac{1}{2}\left\| \begin{matrix} w \\ b \end{matrix} \right\|^2 + \frac{C}{2}\sum\limits_{j\in J^+}\xi_j^2 + C\sum\limits_{j\in J^-}\xi_j \\ & \xi_j = 1 - (w^T x_j + b) \quad j \in J^+ \\ & \xi_j \geq 1 + (w^T x_j + b) \quad j \in J^- \\ & \xi_j \geq 0 \quad j \in J^-, \end{cases} \tag{2}$$

by varying the sets J^+ and J^-, which contain the indexes of the instances currently considered positive and negative, respectively. The positive parameter C tunes the weight between the maximization of the margin, obtained by minimizing the Euclidean norm of (w, b), and the minimization of the misclassification errors of the instances x_js. At the initialization step, J^+ contains the indexes of all the instances of the positive bags, while J^- contains the indexes of all the instances of the negative bags. Once an optimal solution, say (w^*, b^*, ξ^*), to problem (2) has been computed, the two sets J^+ and J^- are updated in the following way:

$$J^+ := J^+ \setminus \bar{J} \qquad \text{and} \qquad J^- := J^- \cup \bar{J}$$

where

$$\bar{J} = \{j \in J^+ \setminus J^* \mid w^{*T} x_j + b^* \leq -1\},$$

with

$$J^* = \{j_i^*,\ i = 1, \ldots, m \mid w^{*T} x_{j_i^*} + b^* \leq -1\}$$

and

$$j_i^* \triangleq \arg \max_{j \in (J_i^+ \cap J^+)} \{w^{*T} x_j + b^*\}$$

Some comments on the updating of the sets J^+ and J^- are in order. A particular role in the definition of the set $\bar{J}$ is played by the set J^*, introduced for taking into account the satisfaction of the standard MIL assumption, stating that, for each positive bag, at least one instance must be positive. At the current iteration, the set J^* is the index set (subset of J^+) corresponding to the instances closest, for each positive bag, to the current hyperplane $H(w^*, b^*)$ and strictly lying in the negative side with respect to it. If an index, say $j_i^* \in J^*$, corresponding to one of such instances entered the set J^-, all the instances of the i-th positive bag would be considered negative by problem (2), favouring the violation of the standard MIL assumption. This is the reason why the indexes of J^* are prevented from entering the set J^-: in this way, for each positive bag, at least an index corresponding to one of its instances is guaranteed to be inside J^+.

4 NUMERICAL EXPERIMENTS

Algorithm mi-SPSVM, summarized in the previous section, has been tested on a random sample of eye fundus images drawn from the public dataset Messidor, available at the web page https://www.adcis.net/en/third-party/messidor/. It has been run on a Windows 11 system with a 2.21 GHz processor, using the same Matlab implementation adopted in [6, 32].

The Messidor database [10] consists of 1200 color images of the back of the eye (the posterior pole), taken by three ophthalmology departments using a color video camera mounted on a Topcon TRC NW6 retinograph. The camera had a 45-degree field of view, and the images have been captured with 8 bits per color plane at various resolutions (1440×960, 2240×1488, or 2304×1536 pixels). All the images have been used for clinical diagnoses, and patient identifying information has been removed to ensure privacy. In correspondence to each image, two types of diagnoses have been provided by medical experts: a grade for retinopathy and a risk assessment for macular edema.

Out of the 1200 images in the database, 800 were taken with pupil dilation using Tropicamide at 0.5%, while 400 were taken without dilation. The images are divided into three sets, one for each ophthalmology department, and each set is further divided into four zipped subsets containing 100 images in TIFF format and an Excel file with medical diagnoses for each image.

In our numerical experiments, we have performed the classification of eye fundus of people affected by severe diabetic retinopathy (positive bags, Figure 6) against those ones of healthy people (negative bags, Figure 7).

Regarding the segmentation process, we have used a procedure similar to that one described in [6, 32]. The resolution of each image has been reduced to 128×128 pixels, and the pixels have been grouped into square subregions, called "blobs." This has allowed us to represent each image as a bag, where each blob corresponds to an instance of the bag. For each instance (blob), we have considered the following features:

- the average of the RGB (red, green, blue) intensities of the blob extracted from the eye fundus: 3 features;
- the variance of the RGB (red, green, blue) intensities of the blob extracted from the eye fundus: 3 features;
- the differences between the average of the RGB intensities of the blob and the same corresponding quantities computed for the adjacent blobs (up, down, left, right): 12 features;
- the differences between the variance of the RGB intensities of the blob and the same corresponding quantities computed for the adjacent blobs (up, down, left, right): 12 features.

Since we have not considered the blobs along the frame of each image, we have come out with a data set where each image is a bag

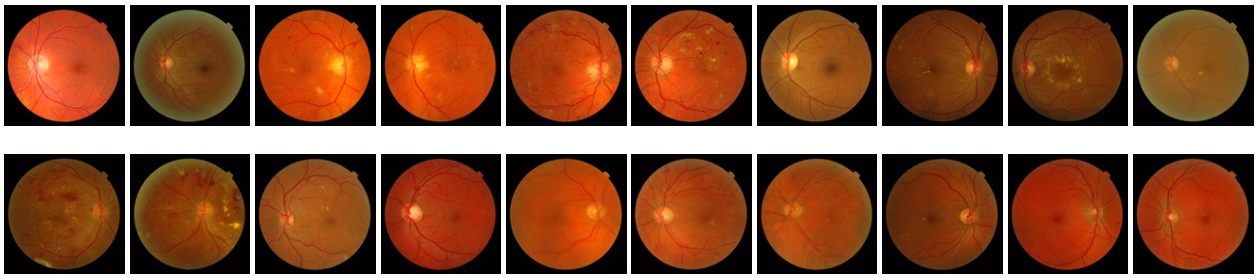


Figure 6: Examples of eye fundus color images of people affected by severe Diabetic Retinopathy

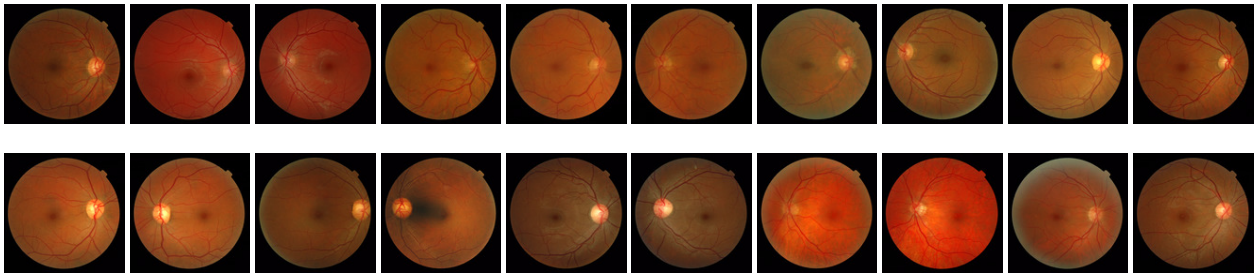


Figure 7: Examples of eye fundus color images of healthy people

characterized by sixteen instances (the blobs) and 30 features for each instance.

In order to consider in our tests different sizes of the testing and training sets, we have used three different validation protocols: the 5-fold cross-validation (5-CV), the 10-fold cross-validation (10-CV) and the Leave-One-Out (LOO) validation. As for the optimal computation of the tuning parameter C characterizing model (2), we have adopted a belevel approach of the type used in [3]: such approach splits the data into two levels and performs the computation of C using only the second level data, since the first level ones are used to train and to test the classifier.

The average testing results we have obtained (in terms of accuracy, sensitivity, specificity, F-score and CPU time) are reported in Tables 2, 3 and 4, comparing against Algorithm MIL-RL [4], which is another MIL technique preliminarily used in [26] for diabetic retinopathy images classification. For the sake of completeness, we also report the results obtained by running the standard SVM technique, for which we have used the linear and the Gaussian kernels (columns SVM and SVM-RBF, respectively). For each table and for each type of performance metric, the best results are reported in bold.

Looking at Tables 2, 3 and 4, we can observe that algorithm mi-SPSVM exhibits the best performance not only in terms of accuracy, but also in terms of sensitivity and F-score, which are the most significant parameters in medical diagnosis since they express the capability of the classifier to correctly individuate positive patients. Moreover, in terms of CPU time, even if the winner appears the SVM with RBF kernel, it also evident that mi-SPSVM is much faster than MIL-RL.

	5-CV			
	mi-SPSVM	MIL-RL	SVM	SVM-RBF
Accuracy (%)	**73.50**	72.00	54.00	67.50
Sensitivity (%)	**78.79**	60.41	47.99	64.40
Specificity (%)	68.35	82.66	60.82	**71.30**
F-score (%)	**74.54**	67.36	50.52	66.49
CPU time (secs)	0.27	3.02	1.99	**0.04**

Table 2: 100 healthy and 100 diabetic retinopathy eye fundus images: 5-CV average testing values

	10-CV			
	mi-SPSVM	MIL-RL	SVM	SVM-RBF
Accuracy (%)	**74.50**	72.00	53.50	68.50
Sensitivity (%)	**77.28**	60.65	56.25	65.32
Specificity (%)	70.48	82.18	51.11	**72.02**
F-score (%)	**73.40**	66.32	53.12	66.71
CPU time (secs)	0.38	3.17	2.03	**0.01**

Table 3: 100 healthy and 100 diabetic retinopathy eye fundus images: 10-CV average testing values

5 CONCLUSIONS AND FUTURE WORK

In this work, we have presented some preliminary numerical results obtained from the classification of severe diabetic retinopathy fundus images against normal ones images, by means of a MIL algorithm. These results appear promising, especially if we take into

	LOO			
	mi-SPSVM	MIL-RL	SVM	SVM-RBF
Accuracy (%)	**73.50**	70.50	59.00	70.50
Sensitivity (%)	**76.00**	56.00	49.00	70.00
Specificity (%)	**71.00**	85.00	69.00	**71.00**
F-score (%)	**74.15**	65.50	54.44	70.35
CPU time (secs)	0.60	4.71	3.20	**0.01**

Table 4: 100 healthy and 100 diabetic retinopathy eye fundus images: LOO average testing values

account that no preprocessing phase has been performed. Moreover, the adopted MIL technique appears appealing also in terms of computational efficiency since the separation hyperplane is always obtained in less than one second.

Future research could consist in appropriately preprocessing the images and in considering additional features to be exploited in the classification process. The obtained classification performances can also give important results using ensemble learning configurations. In this sense, what is presented in this work is preliminary to the creation of hybrid solutions, based on the use of DL approaches for the preprocessing and segmentation steps of fundus images and on the adoption of advanced ML algorithms for the classification phase.

REFERENCES

[1] J. Amores. 2013. Multiple instance classification: Review, taxonomy and comparative study. *Artificial Intelligence* 201 (2013), 81–105.

[2] A. Astorino, M. Avolio, and A. Fuduli. 2022. A maximum-margin multisphere approach for binary Multiple Instance Learning. *European Journal of Operational Research* 299, 2 (2022), 642–652.

[3] A. Astorino and A. Fuduli. 2016. The proximal trajectory algorithm in SVM cross validation. *IEEE Transactions on Neural Networks and Learning Systems* 27, 5 (2016), 966–977.

[4] A. Astorino, A. Fuduli, and M. Gaudioso. 2019. A Lagrangian Relaxation Approach for Binary Multiple Instance Classification. *IEEE Transactions on Neural Networks and Learning Systems* 30, 9 (2019), 2662 – 2671.

[5] M. Avolio and A. Fuduli. 2021. A semiproximal support vector machine approach for binary multiple instance learning. *IEEE Transactions on Neural Networks and Learning Systems* 32, 8 (2021), 3566–3577.

[6] M. Avolio, A. Fuduli, E. Vocaturo, and E. Zumpano. 2022. Multiple Instance Learning for Viral Pneumonia Chest X-ray Classification. In *CEUR Workshop Proceedings*, Vol. 3194. 359–366.

[7] C. Bergeron, G. Moore, J. Zaretzki, C. Breneman, and K. Bennett. 2012. Fast Bundle Algorithm for Multiple Instance Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 34, 6 (2012), 1068 – 1079.

[8] M.A. Carbonneau, V. Cheplygina, E. Granger, and G. Gagnon. 2018. Multiple instance learning: a survey of problem characteristics and applications. *Pattern Recognition* 77 (2018), 329 – 353.

[9] L. Caroprese, E. Vocaturo, and E. Zumpano. 2022. Argumentation approaches for explanaible AI in medical informatics. *Intelligent Systems with Applications* 16 (2022), 200109.

[10] E. Decencière, X. Zhang, G. Cazuguel, B. Lay, B. Cochener, C. Trone, P. Gain, R. Ordonez, P. Massin, A. Erginay, et al. 2014. Feedback on a publicly distributed image database: the Messidor database. *Image Analysis & Stereology* 33, 3 (2014), 231–234.

[11] T.G. Dietterich, R.H. Lathrop, and T. Lozano-Pérez. 1997. Solving the Multiple Instance Problem with Axis-parallel Rectangles. *Artificial Intelligence* 89, 1-2 (1997), 31–71.

[12] Xibin Dong, Zhiwen Yu, Wenming Cao, Yifan Shi, and Qianli Ma. 2020. A survey on ensemble learning. *Frontiers of Computer Science* 14 (2020), 241–258.

[13] A. Fuduli, M. Gaudioso, W. Khalaf, and E. Vocaturo. 2022. A heuristic approach for multiple instance learning by linear separation. *Soft Comput.* 26, 7 (2022), 3361–3368.

[14] G. Fung and O.L. Mangasarian. 2001. Proximal support vector machine classifiers. In *Proceedings KDD-2001: Knowledge discovery and data mining*, F. Provost and R. Srikant (Eds.). ACM, 77–86.

[15] M. Gaudioso, G. Giallombardo, G. Miglionico, and E. Vocaturo. 2020. Classification in the multiple instance learning framework via spherical separation. *Soft Computing* 24, 7 (2020), 5071–5077.

[16] F. Herrera, S. Ventura, R. Bello, C. Cornelis, A. Zafra, D. Sánchez-Tarragó, and S. Vluymans. 2016. *Multiple instance learning: Foundations and algorithms.* Springer International Publishing.

[17] E. Imani, H.-R. Pourreza, and T. Banaee. 2015. Fully automated diabetic retinopathy screening using morphological component analysis. *Computerized medical imaging and graphics* 43 (2015), 78–88.

[18] Z. Li, S. Keel, C. Liu, Y. He, W. Meng, J. Scheetz, P.Y. Lee, J. Shaw, D. Ting, T.Y. Wong, et al. 2018. An automated grading system for detection of vision-threatening referable diabetic retinopathy on the basis of color fundus photographs. *Diabetes care* 41, 12 (2018), 2509–2516.

[19] O.L. Mangasarian and E.W. Wild. 2008. Multiple instance classification via successive linear programming. *Journal of Optimization Theory and Applications* 137, 3 (2008), 555–568.

[20] K. Ogurtsova, L. Guariguata, N.C. Barengo, P.L.-D. Ruiz, J.W. Sacre, S. Karuranga, H. Sun, E.J. Boyko, and D.J. Magliano. 2022. IDF diabetes Atlas: Global estimates of undiagnosed diabetes in adults for 2021. *Diabetes research and clinical practice* 183 (2022), 109118.

[21] G. Quellec, G. Cazuguel, B. Cochener, and M. Lamard. 2017. Multiple-Instance Learning for Medical Image and Video Analysis. *IEEE Reviews in Biomedical Engineering* 10 (2017), 213–234.

[22] G. Rani, A. Misra, V.S. Dhaka, D. Buddhi, R.K. Sharma, E. Zumpano, and E. Vocaturo. 2022. A multi-modal bone suppression, lung segmentation, and classification approach for accurate COVID-19 detection using chest radiographs. *Intelligent Systems with Applications* 16 (2022), 200148.

[23] G. Rani, A. Misra, V.S. Dhaka, E. Zumpano, and E. Vocaturo. 2022. Spatial feature and resolution maximization GAN for bone suppression in chest radiographs. *Comput. Methods Programs Biomed.* 224 (2022), 107024.

[24] G. Rani, P. Thakkar, A. Verma, V. Mehta, R. Chavan, V.S. Dhaka, R.K. Sharma, E. Vocaturo, and E. Zumpano. 2022. KUB-UNet: Segmentation of Organs of Urinary System from a KUB X-ray Image. *Comput. Methods Programs Biomed.* 224 (2022), 107031.

[25] D.S.W. Ting, C. Y.-L. Cheung, G. Lim, G.S.W. Tan, N.D. Quang, A. Gan, H. Hamzah, R. Garcia-Franco, I.Y. San Yeo, S.Y. Lee, et al. 2017. Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *Jama* 318, 22 (2017), 2211–2223.

[26] E. Vocaturo and E. Zumpano. 2021. Diabetic Retinopathy Images Classification via Multiple Instance Learning. In *2021 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*. 143–148.

[27] E. Vocaturo, E. Zumpano, G. Giallombardo, and G. Miglionico. 2020. DC-SMIL: a multiple instance learning solution via spherical separation for automated detection of displastyc nevi. In *IDEAS 2020: 24th International Database Engineering & Applications Symposium, Seoul, Republic of Korea, August 12-14, 2020*, Bipin C. Desai and Wan-Sup Cho (Eds.). ACM, 4:1–4:9.

[28] E. Vocaturo, E. Zumpano, and P. Veltri. 2018. Features for melanoma lesions characterization in computer vision systems. In *2018 9th International Conference on Information, Intelligence, Systems and Applications (IISA)*. IEEE, 1–8.

[29] H. Yazid, H. Arof, and H.M. Isa. 2012. Automated identification of exudates and optic disc based on inverse surface thresholding. *Journal of medical systems* 36 (2012), 1997–2004.

[30] W.L. Yun, U.R. Acharya, Y.V. Venkatesh, C. Chee, L.C. Min, and E.Y.K. Ng. 2008. Identification of different stages of diabetic retinopathy using retinal optical images. *Information sciences* 178, 1 (2008), 106–121.

[31] Z.-H. Zhou. 2021. *Ensemble learning.* Springer.

[32] E. Zumpano, A. Fuduli, E. Vocaturo, and M. Avolio. 2021. Viral Pneumonia Images Classification by Multiple Instance Learning: Preliminary Results. In *Proceedings of the 25th International Database Engineering & Applications Symposium* (Montreal, QC, Canada) *(IDEAS '21)*. Association for Computing Machinery, New York, NY, USA, 292–296.

[33] E. Zumpano, P. Iaquinta, L. Caroprese, F. Dattola, G. Tradigo, P. Veltri, and E. Vocaturo. 2019. Simpatico 3d mobile for diagnostic procedures. In *Proceedings of the 21st International Conference on Information Integration and Web-based Applications & Services*. 468–472.